1 Research Question and Data

We investigate whether systems can answer multilingual questions from English contexts while abstaining when the context does not support an answer. We separate three sources of apparent success: predictable question forms, answerability classification, and extraction of the correct span. A Korean-to-English translation experiment tests whether changing the question language improves QA under controlled conditions.

We use the Arabic (ar), Korean (ko), and Telugu (te) subset of `coastalcph/tydi_xor_rc` (CoAStaL NLP Group, 2026): 6,335 training and 1,155 validation examples. Contexts are English; questions retain their original language. Table 1 shows substantial label shift: unanswerability rises from 5.73% in training to 14.20% in validation, particularly for Telugu. This motivates class-balanced metrics alongside accuracy.

Split	Lang.	n	U%	Q: m/IQR	C: m/IQR
Train	ar	2558	9.97	6/3	89/68
	ko	2422	2.60	5/2	84/63
	te	1355	3.32	6/2	80/59
Valid.	ar	415	12.53	6/3	87/63
	ko	356	5.34	5/2	84/71
	te	384	24.22	6/2	100/63

Table 1: Subset statistics. U: unanswerable; Q/C: question/context lengths in whitespace tokens.

No required field contains missing values or blank strings; only optional `answer_inlang` is missing (7,340/7,490 records), and it is not used as input. Korean training contains ten duplicate pairs (20 participating rows); manual inspection confirmed agreement across all fields. We retain the ten extra rows (0.16% of training) to preserve the supplied split. Other within-split/language duplicate counts are zero. All 6,963 answerable spans exactly match their context slices. Separately, character-offset tokenisation and BIO conversion recover the original start/text for five sampled answerable examples; five unanswerable examples produce all-O labels and an empty span. These ten successful round trips verify the sampled cases, not exhaustive boundary coverage.

Tokenisation sensitivity. Whitespace counts let isolated question marks occupy top-five positions. Replacing Unicode punctuation with spaces before counting merges punctuation variants without joining neighbouring words. The revised top five, with counts and contextual English glosses, are: **ar**: في (in, 593), من (from/who, 587), متى (when, 536), ما (what, 443), هو (he/it, 350); **ko**: 가장 (most, 527), 무엇인가 (what is, 497), 언제 (when, 336), 몇 (how many, 234), 어디인가 (where is, 228); **te**: ఎవరు (who, 274), ఏది (which one, 192), ఎన్ని (how many, 165), ఎప్పుడు (when, 154), ఏ (which, 144). Telugu "who" rises from 186 to 274 and "which one" enters the ranking, supporting the punctuation-fragmentation explanation. This diagnostic does not change the whitespace protocol used for length statistics or language modelling.

2 Methods and Evaluation Protocol

Question language model. Our Korean bigram model uses all 2,422 training questions, whitespace tokenisation, and add-k smoothing with $k = 0.1$. Tokens occurring fewer than twice in training, and unseen validation tokens, map to `<UNK>`. The 1,126-token prediction vocabulary includes `<UNK>` and `<EOS>`; `<BOS>` supplies history only. For history h and next token w,

$$P(w \mid h) = \frac{C(h, w) + 0.1}{C(h) + 0.1|V|}. \quad (1)$$

Perplexity pools negative log probabilities over all scored validation tokens, including one `<EOS>` per question; it is not the average of question-level perplexities. Model comparisons require identical tokenisation, vocabulary, unknown handling, and boundary scoring.

Answerability models. The majority baseline always predicts answerable, as determined from training. Logistic regression (LR) combines separate question/context TF-IDF blocks with four standardised numerical features: both token lengths, shared distinct-token count, and question-token overlap ratio. Lowercase whitespace tokenisation and minimum document frequency two yield 31,183 features. Vocabulary, IDF, and scaling are fitted on training only; answers and offsets are excluded. LR uses balanced class weights and $C = 1$.

The recurrent classifier shares a randomly initialised, trainable embedding between questions and contexts, encodes them with separate BiLSTMs, and concatenates their final forward/backward states before a nonlinear classifier. Concatenation provides a simple joint decision input; shared embeddings reuse parameters and identical token entries. Neither establishes cross-language

alignment. There is no attention or explicit token-to-token interaction. Class-weighted cross-entropy is minimised, and the best of ten epochs is selected by validation macro-F1. Both classifiers use a 0.5 decision threshold in the runs reported here; BiLSTM results are from seed 42.

Integrated QA and abstention. We fine-tune XLM-R-base (Conneau et al., 2020) jointly on the three languages using question–context inputs and start/end prediction heads. Subword windows have maximum length 384 and context overlap 128. A window without the complete gold answer receives CLS/CLS labels, including windows from answerable examples. Decoding searches context-only spans of at most 30 subwords and slices the original context. Across windows, let s_{null} be the minimum CLS start-plus-end logit and s_{span} the maximum valid span score. We return an empty answer when

$$d = s_{\text{null}} - s_{\text{span}} > \tau. \tag{2}$$

These are unbounded logit differences, not probabilities. At every epoch, one global threshold maximises overall validation QA token-F1; the best epoch and its threshold are saved together. Six-epoch runs use seeds 42, 43, and 44. Gold unanswerable answers are empty for SQuAD-v2-style scoring.

We report normalised exact match (EM), token-overlap F1, and binary answerability metrics, overall and by language. QA scores use 0–100; classification scores use 0–1. Validation serves model/threshold selection, so these are development results, not independent test estimates. Neural QA summaries use mean ± sample standard deviation over three seeds.

3 Validation Results and Diagnostic Analysis

3.1 Question Predictability Is Not Understanding

Bigram validation perplexity is 24.3639. Of 1,729 text tokens, 662 (38.29%) map to <UNK>; scoring includes another 356 <EOS> tokens. Our hypothesis is that frequent local transitions lower perplexity, whereas unseen transitions raise it. Table 2 illustrates both effects using actual questions and GPT-assisted English translations.

The first question becomes two <UNK> tokens followed by 무엇인가?; two semantically unrelated questions collapse to the same sequence and receive identical PPL 3.8087. The all-<UNK> Rome question has PPL 3.8906: its repeated <UNK> transition occurs 992 times and has probability 0.292. In contrast, the Bible question contains no unknown tokens, but its transition counts are only 3, 0, 1, and 2; the unseen transition has probability 0.000865. These cases support the local-frequency hypothesis and show why low perplexity can reflect information loss, rather than semantic competence. These selected cases do not constitute a statistical test.

Question / English translation	PPL
육파철학의 이름들은 무엇인가? What are the names of the six schools of philosophy?	3.81
로마에 에트루리아인 왕조는 몇년간 로마를 지배했나요? For how many years did the Etruscan dynasty rule Rome?	3.89
성경은 언제 쓰여졌는가? When was the Bible written?	380.83

Table 2: Low/high-perplexity validation examples.

	Overall	ar	ko	te
Majority mF1	.462	.467	.486	.431
LR mF1	.813	.957	.846	.688
BiLSTM mF1	.808	.957	.846	.669
QA mF1	.868	.929	.809	.829
QA EM	57.11	57.67	59.27	54.51
QA token-F1	66.36	67.02	70.03	62.25

Table 3: Validation comparison. QA rows average seeds 42/43/44; classifier rows use the reported fixed configuration (BiLSTM seed 42). mF1: answerability macro-F1.

3.2 Classification and Integrated QA

The majority baseline achieves 85.80% accuracy but only 0.4618 macro-F1 and zero unanswerable recall/F1. LR reaches 0.8134 macro-F1, slightly exceeding BiLSTM's 0.8076 despite lower accuracy (91.26% versus 91.43%). Unanswerable precision/recall/F1 are .7114/.6463/.6773 for LR and .7481/.5976/.6644 for BiLSTM; answerable values are .9423/.9566/.9494 and .9355/.9667/.9509. Thus BiLSTM's accuracy gain accompanies poorer minority-class recall. Its best epoch is 3; training loss subsequently falls while validation loss generally rises, consistent with overfitting.

Both classifiers make eight Arabic errors, on the same examples. Telugu contributes 83/101

LR errors and 81/99 BiLSTM errors. Its larger validation unanswerable share and longer contexts are plausible difficulties, not demonstrated causes. LR's strongest negative question feature is Arabic هل, a yes/no particle (coefficient -13.73); all ten strongest positive features are question tokens. These associations suggest question-form shortcuts, but coefficients alone do not establish causality or show that context is unnecessary.

Integrated QA reaches EM 57.11 ± 1.18, token-F1 66.36 ± 1.24, and answerability macro-F1 $.8678 \pm .0222$. Its mean macro-F1 exceeds LR by .0544, particularly improving Telugu, while Arabic/Korean classifier scores remain higher (Table 3). This comparison also changes pretraining, supervision, and threshold selection, so the gain cannot be attributed to architecture alone. Korean has the highest QA F1 but the lowest QA answerability macro-F1: aggregate span scoring and balanced classification measure different behaviour. Only 19 Korean examples are unanswerable. Mean unanswerable precision is .6025, below LR's .8000, despite higher recall (.6842 versus .6316). Thus the metrics expose a rejection trade-off; class imbalance alone is not an explanation for low macro-F1.

Across QA seeds, unanswerable precision/recall/F1 average .7686/.7785/.7734, versus .9633/.9613/.9623 for answerable examples. Unanswerable F1 has higher seed variability (SD .0386 versus .0059), reinforcing the need for class-specific evaluation.

What explains the EM–F1 gap? For the frozen seed-42 model, EM is 57.3160 and F1 is 66.3898. Of 1,155 predictions, 662 are exact matches, 217 have partial token overlap, and 276 have zero overlap or incorrect abstention. No prediction has F1=100 with EM=0. The 217 partial matches (18.79%) therefore explain the entire 9.0738-point gap; their mean F1 is 48.296. Five inspected cases span all languages and different F1 levels: two answers are too long, two too short, and one is semantically wrong despite lexical overlap. For example, predicting "1 May 1944" against "1944" gives F1=50; an RNA/DNA answer sharing only "in" receives 6.06. Token overlap is thus not necessarily partial semantic correctness. Five selected cases do not establish population frequencies for these causes.

Metric	Korean	Translated
QA EM	54.12 ± 0.16	57.87 ± 1.49
QA token-F1	63.99 ± 0.41	67.28 ± 1.56
Answerability mF1	$.7689 \pm .0310$	$.7896 \pm .0584$
Unanswerable F1	$.5636 \pm .0553$	$.6031 \pm .1076$

Table 4: Controlled translation experiment: mean $\pm$ sample SD over three matched seeds.

4 Translation Extension and Limitations

Hypothesis and controlled comparison. Translating Korean questions into English should improve QA token-F1. To isolate question-language transformation, we retrain the same XLM-R-base QA architecture on Korean-only data in both conditions (2,422 training, 356 validation examples). GPT translates questions in both splits; contexts, gold spans, labels, and record order remain unchanged, with no manual editing or selection. The two conditions therefore evaluate the same underlying examples and answers. They share settings, seeds 42/43/44, and the validation threshold-selection rule, not necessarily the numerical threshold. The three-language model is not the controlled baseline because its training data differ.

Translation improves mean token-F1 by 3.29 points and EM by 3.75 points (Table 4); token-F1 improves for every matched seed, by 3.19, 4.90, and 1.77 points. This supports the hypothesis within this setup, without establishing statistical significance or transfer to other languages. Answerability macro-F1 decreases for seeds 42 and 43 and increases for 44; its higher mean is not a consistent abstention improvement. Translation may reduce the question–context language mismatch, but can also alter phrasing or meaning; the experiment does not isolate these mechanisms.

Limits and conclusion. The QA preprocessing covers every answerable training example in at least one window, but 49 examples have covered answers exceeding the 30-subword decoding cap. Validation reuse, small minority classes, and three seeds limit generalisation claims. The random-embedding BiLSTM has no explicit cross-language alignment, and the LM loses substantial lexical information through `<UNK>`. Integrated QA improves aggregate answerability; translation consistently improves answer-text scores, but not abstention. These outcomes require separate evaluation.

References

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